# Lubrication System

CONTENTS OF THIS SECTION


| Subject | Page | Subject | Page |
|---|---|---|---|
| Circulation and Distribution | 72 | Oil Cooler | 79 |
| Engine Oil Recommendations | 78 | Lubricating Oil Pump | 82 |
| Oil Filters and Strainers | 79 | Oil Pressure Regulator Valve | 87 |


## CIRCULATION AND DISTRIBUTION

The full-flow lubrication system, used on all latest engines, is schematically shown in figure 1 and consists of oil pump, oil filter, oil cooler, and by-pass valves around filter and cooler.

The by-pass lubrication system, schematically shown in figure 2, has been superseded by the full-flow type and consisted of oil pump, oil filter, oil cooler, oil strainer (coach only) and cooler by-pass valve.

The oil cooler adapter has a spring loaded by-pass valve which assures continual lubrication even through oil cooler fails to function. Should the oil pressure at cooler inlet become 40 pounds greater than at cooler outlet, the by-pass valve in cooler adapter opens, and oil is then by-passed directly from oil pump to oil gallery. However, under these conditions, the oil is not cooled.

The gallery oil pressure is limited to 45 pounds per square inch by a spring loaded pressure regulator valve at the end of oil gallery. While a part of oil gallery, this regulator is actually located in crankcase, and is integral with pump discharge line.

Continuous filter action is provided by by-passing a portion of the oil through a filter from whence filtered oil is returned to crankcase.

Oil from cooler is conducted through a vertical passage to a longitudinal main gallery on blower side of crankcase. As shown in figures 3 and 4 this gallery distributes the oil, under pressure, to main bearings, and to a horizontal, transverse passage at each end of cylinder block. From these two horizontal passages, two vertical bores at each end of cylinder block carry the oil to end bearings of camshaft and balance shaft, as well as to oil passage in camshaft, which in turn conducts the oil to camshaft intermediate bearings.

Oil for lubrication of connecting rod bearings, piston pins, and the cooling of piston head, is provided through drilled crankshaft from the adjacent forward main bearings. The gear train is lubricated by overflow of oil from camshaft pocket through a passage into gear train cover. Oil spills into the gear cover from camshaft, balancer shaft, and idler gear bearings. The blower drive gear bearing is lubricated through an external pipe from rear horizontal oil passage of cylinder block.

A second longitudinal gallery is arranged on camshaft side of cylinder head and supplied with oil from one of the vertical bores at each end of cylinder block. Oil from this gallery enters hollow rocker arm shafts through rocker shaft brackets, hollow capscrews, and lubricates rocker arm bearings and push-rod clevis bearings.

Excess oil from rocker arms lubricates valve ends and push rods, and drains to cam pockets in cylinder head from which cams are lubricated. As shown in figure 5, after reaching a certain level, this oil overflows through two holes at each end of blower housing, providing lubrication for blower drive gears at rear end and governor drive gear at the front. A dam in blower housing cover maintains an oil level which submerges teeth of the lower blower rotor timing gear. A slinger on opposite end of lower rotor throws oil into governor weight assembly. Surplus oil passes from blower to oil pan through drilled holes in cylinder block.

## CRANKCASE VENTILATION

Under normal operating conditions, harmful vapors may collect in crankcase and valve compartment. These vapors may, under certain conditions, cause etching of highly finished surfaces and dilution of oil unless withdrawn. The crankcase and valve compartment are continually cleaned of harmful vapors by a breather pipe attached to governor upper housing and venting to atmosphere.